Statistical Analysis of MetroPT-3 Compressor Sensor Behavior

Peyman Mohammad Hassan
Master's in Artificial Intelligence — Statistical Analysis Project
Independent research using the MetroPT-3 public sensor dataset

Overview

This project examines recorded sensor measurements from a metro train air-production compressor using reproducible descriptive and inferential statistics. The goal is to characterize sensor distributions and evaluate whether panel pressure differs between operating-signal states within the same day. The findings support the analytical foundations of QIAIP ERP condition monitoring; they are not validated fault predictions or evidence of actual maintenance work orders. Dataset: MetroPT-3, https://archive.ics.uci.edu/dataset/791/metropt%2B3%2B.

Dataset Description

The MetroPT-3 dataset originates from the UCI Machine Learning Repository and records pressure, motor current, temperature and digital-control signals of a railway compressor (Davari et al., 2021). The official repository contains 1,516,948 records and 15 measured features. This project uses 30,339 authentic sampled rows (every 50th original row) and 17 CSV columns, including source-row index and timestamp. The sample spans February 1 through September 1, 2020, with 212 observed dates. Measurements are from a single system, not from independent compressors. Dataset URL: https://archive.ics.uci.edu/dataset/791/metropt%2B3%2B

Workflow and Data Preparation

Pandas loads the accompanying CSV; cleaning functions explicitly parse timestamps and numeric fields, check essential measurements and remove duplicate source-row identifiers. Cleaning removed zero rows from this particular sample. The EDA summarizes continuous measurements and operating-state frequencies; four primary graphs examine pressure over time, panel pressure by COMP state, oil-temperature distribution and the relationship between current and pressure. A fifth graph examines the shape of within-day paired pressure differences. The notebook contains an executed two-sided t-test and a sign-flip permutation sensitivity check.

Research Question and Hypotheses

Question: Does mean TP3 pneumatic-panel pressure differ between readings with recorded COMP=1 and COMP=0 when compared within the same calendar day? **Null hypothesis (H0):** The population mean within-day TP3 difference (COMP=1 minus COMP=0) equals zero. **Alternative hypothesis (H1):** The population mean within-day difference does not equal zero. The significance level is 0.05. The COMP channel is an observed operating signal, not a verified equipment-failure indicator.

Methods

Raw compressor readings are temporally repeated observations. Treating individual records as independent experimental replicates would give an unjustified impression of statistical precision (Hurlbert, 1984). Following the longitudinal screening principles of Lusa et al. (2024), the analysis first groups pressure readings by day and observed COMP state, retains dates with both states, and computes one paired difference for each date. A two-sided one-sample t-test of the 205 daily differences is equivalent to a paired comparison of the two daily state-specific means. It assumes the differences are approximately independent across days and that the normal approximation is reasonable; these assumptions are imperfect for this time series. A fixed-seed, 20,000-draw sign-flip sensitivity check provides a complementary exploratory result; it also requires assumptions about the differences.

Results

Across the 30,339 records, 4,940 observations have COMP=0 and 25,399 have COMP=1. Of the 212 days represented, 205 have enough source observations in both states for pairing. The observed mean daily TP3 difference (COMP=1 minus COMP=0) is **0.015469 bar**. The paired test yields **t(204)=0.452670**, **two-sided p=0.651267**, and a **95%**

confidence interval from −0.051908 to 0.082846 bar. The standardized paired effect (Cohen's dz) is approximately **0.032**, small in magnitude. The sign-flip analysis returned an approximate p-value of **0.660017**. Both procedures give little evidence of a mean within-day difference in this sampled dataset; neither establishes that the operational states are identical.

Interpretation for a Non-Technical Audience

Figure 1 documents changes in panel pressure across the available observation dates, while Figure 2 compares the raw distributions of panel pressure by COMP state. Figure 3 shows the spread of oil-temperature readings; Figure 4 illustrates how measured motor current and pressure co-occur across the two COMP states. Figure 5 presents the distribution of within-day pressure differences used in the test. Although many source observations exist, the inferential unit is the day: 205 paired differences, not 30,339 statistically independent experimental subjects. The nonsignificant p-value should be interpreted as insufficient evidence for the stated mean-difference hypothesis, not confirmation of equivalence or sensor reliability.

Limitations and Potential Bias

Systematic every-fiftieth-row sampling makes the notebook practical but may omit short anomalies or bias measured operating-state proportions. Successive days can remain serially dependent, weakening nominal t-test and permutation-test calibration (Lusa et al., 2024; Hurlbert, 1984). Nonrandomized operating states can be confounded by other conditions and equipment scheduling. The research data are obtained from one train compressor, so findings must not be generalized to an entire fleet or to QIAIP customer installations without external verification. UCI separately documents failure-event intervals, but no sampled reading was assigned an invented fault label. Equipment diagnosis and work-order creation would require validated labels, engineering thresholds and human review.

Reproducibility and Version Control

The project includes analysis.ipynb, metropt3_compressor_sample.csv, requirements.txt generated using pip freeze, and a concise README. The notebook was executed from beginning to end using Python 3.12 and the CSV in the same folder; its plots and test results are saved in the file. The original dataset source, deterministic sampling stride (50), and variable meanings are documented. Local Git history records incremental commits and an additional development branch. The independently verified GitHub submission branch preserves the complete analysis and source CSV, with multiple commits showing progress.

Future Integration Reflections

For a supervised machine-learning model, reliable equipment-failure labels and an appropriate future-oriented train/test split would be necessary before claiming predictive skill. For neural-network training, sampling cadence, time-window construction, feature scaling and possible temporal leakage must be audited. Agentic automation in QIAIP could flag unusual sensor behavior and summarize evidence for an engineer, but it could also amplify misleading correlations or label ordinary operations as faults. Proposed maintenance work orders should remain human-approved until their trigger conditions are validated. The responsible AI workflow should preserve original sensor provenance and log every proposed action.

References

Hurlbert, S. H. (1984). Pseudoreplication and the design of ecological field experiments. *Ecological Monographs, 54*(2), 187–211. https://doi.org/10.2307/1942661

Lusa, L., Proust-Lima, C., Schmidt, C. O., Lee, K. J., le Cessie, S., Baillie, M., Lawrence, F., & Huebner, M. (2024). Initial data analysis for longitudinal studies to build a solid foundation for reproducible analysis. *PLOS ONE, 19*(5), e0295726. https://doi.org/10.1371/journal.pone.0295726

Davari, N., Veloso, B., Ribeiro, R., & Gama, J. (2021). *MetroPT-3 dataset* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5VW3R

Veloso, B., Ribeiro, R. P., Gama, J., & Pereira, P. M. (2022). The MetroPT dataset for predictive maintenance. *Scientific Data, 9*, Article 764. https://doi.org/10.1038/s41597-022-01877-3 (Related MetroPT research; not claimed to describe the identical 2020 MetroPT-3 source records.)

Figures from the Executed Statistical Notebook

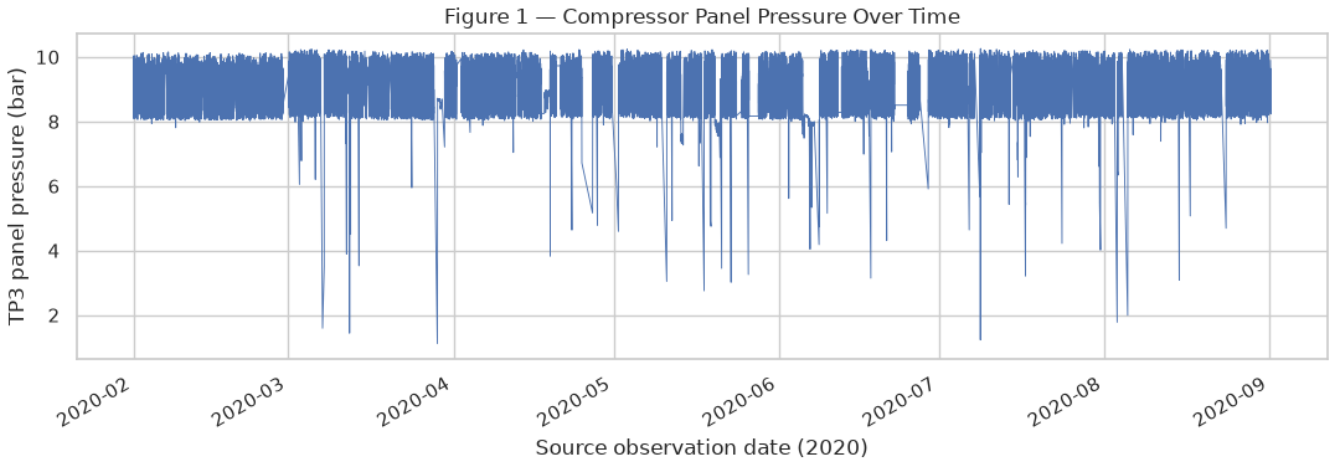


Figure 1. Rendered from executed analysis.ipynb using observed MetroPT-3 sample readings.

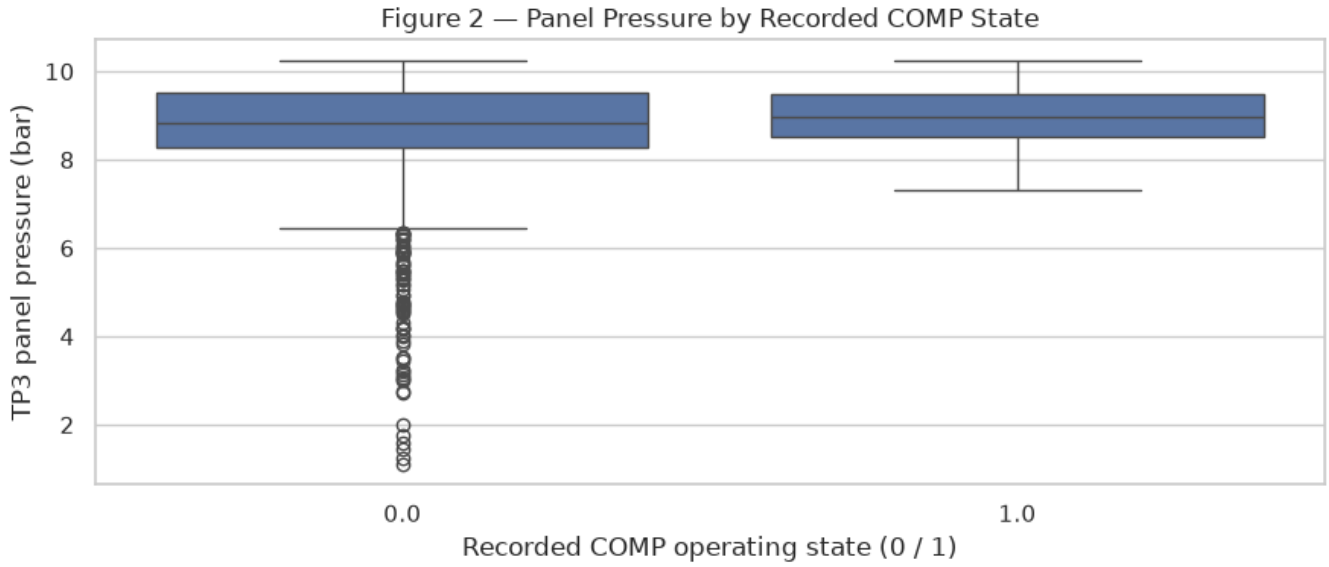


Figure 2. Rendered from executed analysis.ipynb using observed MetroPT-3 sample readings.

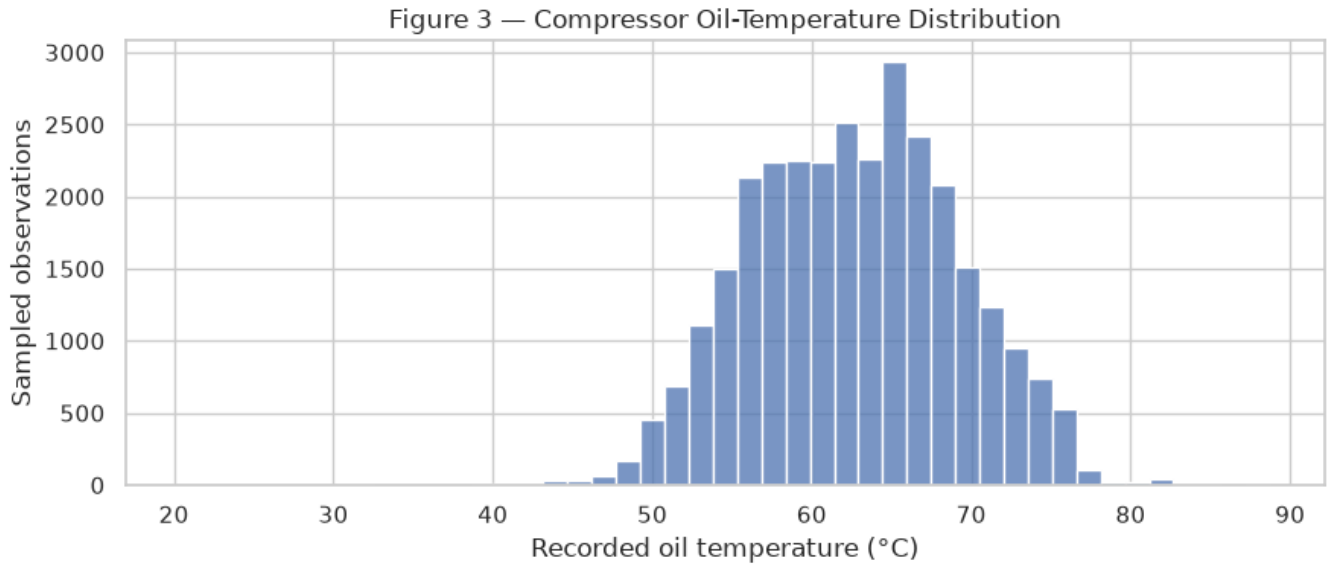


Figure 3. Rendered from executed analysis.ipynb using observed MetroPT-3 sample readings.

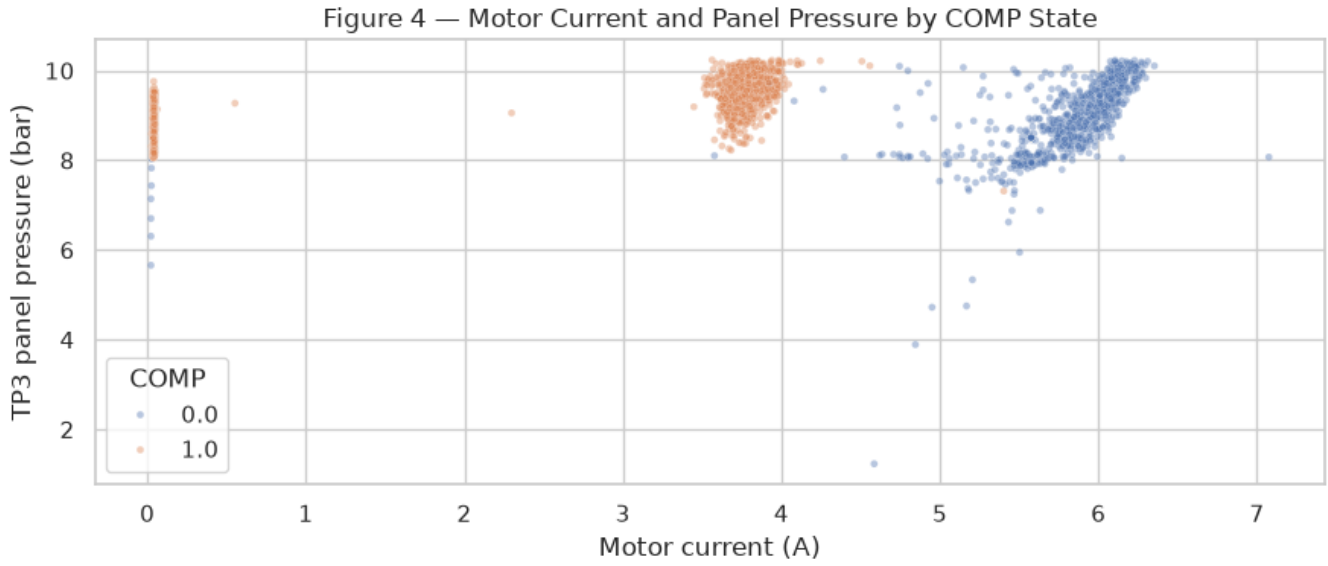


Figure 4. Rendered from executed analysis.ipynb using observed MetroPT-3 sample readings.

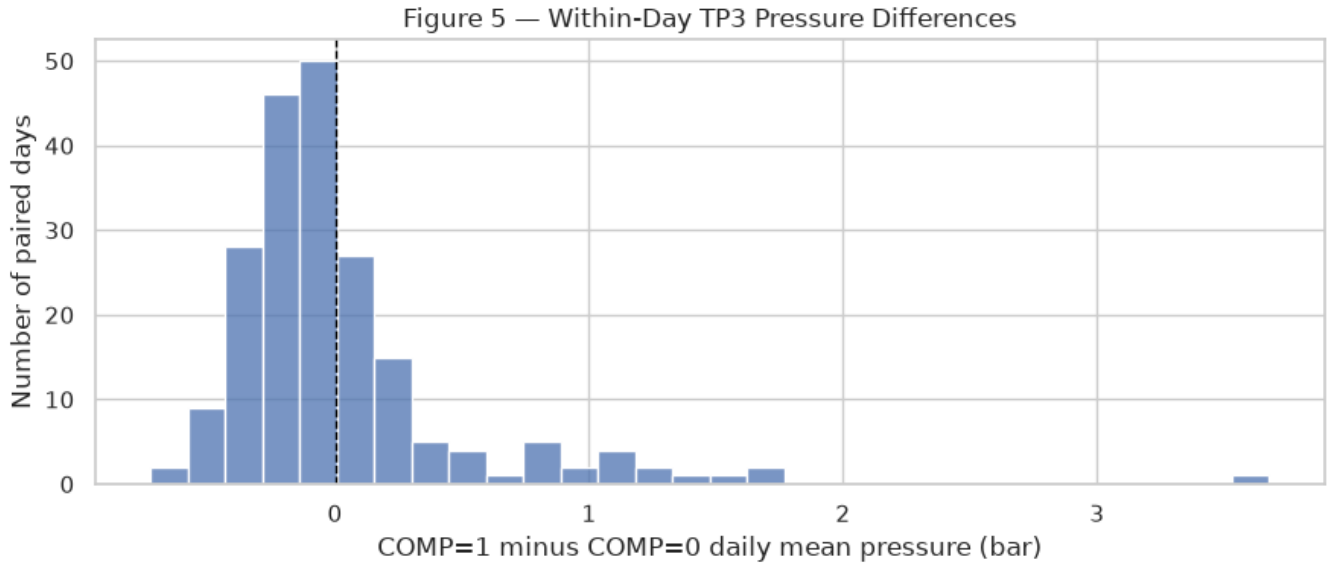


Figure 5. Rendered from executed analysis.ipynb using observed MetroPT-3 sample readings.